

94-34738

REVIEWED

LEGAL COUNSEL

BEFORE THE BOARD OF COUNTY COMMISSIONERS OF DESCHUTES COUNTY, OREGON

94 SEP -1 AM 10:22

An Ordinance Adopting Chapter 15.10 *
Deschutes County Code; Providing *
Outdoor Lighting Control. *

MARY SUE PENHOLLOW
COUNTY CLERK

0137-0091

ORDINANCE NO. 94-024

WHEREAS, the Board affirms the right of citizens in Deschutes County, Oregon to illuminate residential, commercial and public areas with lighting fixtures appropriate to the need while utilizing such illumination in a way that preserves rural and urban vistas and is confined to the property from which it is generated.

WHERE, the Board affirms that citizens of Deschutes County, Oregon have a right to the safety of well-lighted streets and highways and that such illumination by nature cannot be confined to the property from which it is generated. Thus, certain high wattage and low wattage applications for the purpose of highway safety as defined herein are allowed under these provisions.

WHEREAS, public hearings on the proposed ordinance TA-93-19 have been held in conformance with state law before the Deschutes County Planning Commission and Board of County Commissioners for Deschutes County; and

WHEREAS, after notice and hearings as required by law, the Board of County Commissioners has considered the recommendation of the Planning Commission; now therefore,

THE BOARD OF COUNTY COMMISSIONERS OF DESCHUTES COUNTY, OREGON, ORDAINS as follows:

Section 1. That the Deschutes County Code, Title 15, be amended to add Chapter 10, "Outdoor Lighting Control," as described in Exhibit A, attached hereto and by this reference incorporated herein.

Section 2. Codification. The Code Book Review Committee shall have the authority to format the provisions contained herein in a manner that will integrate them into the County Code consistent with the Code Book Review Committee's form and style for ordinance codification. Such codification shall include the authority to make format changes, to make changes in numbering systems and to make such numbering changes consistent with interrelated code sections. In addition, as part of codification of these ordinances, the Code Book Review Committee may insert appropriate legislative history references. Any legislative history references included herein are not adopted as part of the substance of this ordinance, but are included for administrative convenience and as a reference. They may be changed to

1 - Ordinance 94-024 (08/10/94)

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correct errors and to conform to proper style without action of the Board of County Commissioners.

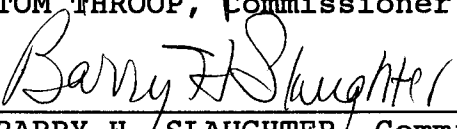
Section 3. Severability. The provisions of this ordinance are severable. If any section, sentence, clause, or phrase of this ordinance or any exhibit thereto is adjudged to be invalid by a court of competent jurisdiction, that decision shall not affect the validity of the remaining portions of this ordinance or any exhibit thereto.

DATED this 31st day of August, 1994.

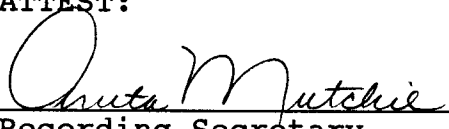
BOARD OF COUNTY COMMISSIONERS
OF DESCHUTES COUNTY, OREGON


NANCY PORE SCHLANGEN, Chair


TOM THROOP, Commissioner


BARRY H. SLAUGHTER, Commissioner

ATTEST:


Anita Mutchie
Recording Secretary

Chapter 15.10**OUTDOOR LIGHTING CONTROL****Sections:**

- 15.10.010 Purpose and intent as relates to residential, commercial and public area lighting.**
- 15.10.020 Purpose and intent as relates to street lighting.**
- 15.10.030 Conformance with applicable codes.**
- 15.10.040 Approved materials and methods of construction or installation/operation.**
- 15.10.050 Definitions.**
- 15.10.060 Requirements for installation of outdoor lighting.**
- 15.10.070 Submission of plans and evidence of compliance with code-subdivision plats.**
- 15.10.080 Shielding.**
- 15.10.090 Prohibitions.**
- 15.10.100 Externally lighted outdoor advertising signs, billboards.**
- 15.10.110 Exemptions.**
- 15.10.120 Violations and penalties.**
- 15.10.130 Violations constitute public nuisance.**
- 15.10.140 Code requirements table for shielding.**
- 15.10.010 Purpose and intent as relates to residential, commercial and public area lighting.**

The purpose of this section is to affirm the right of citizens in Deschutes County, Oregon to illuminate residential, commercial and public areas with lighting fixtures appropriate to the need while utilizing such illumination in a way that preserves rural and urban vistas and is confined to the property from which it is generated.

- 15.10.020 Purpose and intent as relates to street lighting.**

The purpose of this section is to affirm that citizens of Deschutes County, Oregon have a

right to the safety of well-lighted streets and highways and that such illumination by nature cannot be confined to the property from which it is generated. Thus, certain high wattage and low wattage applications for the propose of highway safety as defined below are allowed under these provisions.

- 15.10.030 Conformance with applicable codes.**

All outdoor electrically powered illuminating devices shall be installed in conformance with the provisions of this code, the building code, the electrical code, and the sign code of the jurisdiction. No provision of this ordinance are intended to pre-empt applicable state codes.

- 15.10.040 Approved materials and methods of construction or installation/operation.**

The provisions of this code are not intended to prevent the use of any design, material, or method of installation or operation not specifically prescribed by this code, provided any such alternate has been approved. The building official may approve any such proposed alternate that:

A. Provides an equivalent alternative design that does not exceed 1800 lumens nor project light off-site of the subject lot or parcel.

- 15.10.050 Definitions.**

A. Outdoor light fixtures means outdoor artificial illuminating devices, outdoor fixtures, lamps and other similar devices, permanently installed or portable, used for flood lighting, general illumination or advertisement. Such devices shall include, but are not limited to, search, spot and flood lights for:

1. buildings and structures;
2. recreational areas;
3. parking lot lighting;
4. landscape lighting;
5. billboards and other signs (advertising or other);
6. street lighting
7. product display area lighting;

8. building overhangs and open canopies.
9. holiday lighting.

B. The Community Development Department means the Community Development Department or designated representative(s) for the purposes of the ordinance.

C. Exempt light fixtures means outdoor artificial illuminating devices which are exempted by Section 15.10.110.

D. Individual means any private individual, tenant, lessee, owner or any commercial entity including but not limited to companies, partnerships, joint ventures or corporations.

E. Installed means initial installation of outdoor lighting fixtures following the effective date of this ordinance. Projects with approved construction plans prior to effective date of this ordinance are excluded from compliance with the ordinance in the initial installation only.

F. Shielding for the purpose of this ordinance is provided for a lighting fixture by design of such fixture or by an externally applied device such as a shroud or hood of metal, wood or painted glass that does not allow transmission of light.

1. Fully shielded means outdoor light fixtures shielded or constructed so that light rays emitted by the fixture are projected below the horizontal plane.

2. Partially shielded means shielding so that the edge of the shield is at or below the centerline of the light source or lamp so as to limit light emission above the horizontal plane to 10 percent or less.

3. Directed shielding means shielding by design or external application that directs light downward and limits direct line-of-sight of a fixture's lamp to the property upon which the fixture is installed.

4. Unshielded means light fixtures lacking any means to restrict the emitted light to below the horizontal plane.

G. High intensity discharge lamp sources means high pressure sodium, mercury vapor, metal halide, low pressure sodium, and other similar lamps.

H. Luminous tube lighting means gas-filled tubing which, when subjected to high voltage, becomes luminescent in a color characteristic of the particular gas used, e.g. neon, argon, etc.

15.10.060 Requirements for installation of outdoor lighting.

A. Except as exempted by provisions of this ordinance, as of the date of adoption, the installation of outdoor lighting fixtures shall be subject to the provisions of this ordinance.

15.10.070 Submission of plans and evidence of compliance with code-subdivision plats.

A. All proposed subdivisions and partitions within Deschutes County that include outdoor lighting fixtures or street lighting shall be subject to the provisions of this ordinance.

15.10.080 Shielding.

A. All nonexempt outdoor lighting fixtures shall have shielding as required by the tables set forth in Section 15.10.140.

15.10.090 Prohibitions.

A. Laser Source Light. The use of laser source light or any similar high intensity light for outdoor advertising or entertainment, when projected above the horizontal is prohibited.

B. Searchlights. The operation of searchlights for advertising purposes is prohibited between eleven o'clock P.M. and sunrise the following morning.

C. Recreational Facilities. No outdoor recreational facility, public or private, shall be illuminated after eleven o'clock P.M. except to conclude a specific recreational or sporting event or any other similar activity conducted at or in the facility which was in progress under such illumination prior to eleven o'clock P.M., except that any outdoor recreational facility, public or private, which is illuminated with outdoor lighting fixtures conforming to this code may operate any time with such illumination.

15.10.100 Externally lighted outdoor advertising signs, billboards.

A. All externally lighted advertising signs and billboards will be illuminated by one of the following manners:

1. Top mounted fixtures in which case such fixtures will conform to the shielding requirements as set forth in Section 15.10.140.

2. Bottom mounted fixtures in which case such fixtures shall be shielded either by application of external device or manufactured in such a way that upward and side directed light is confined to an area within 4 (four) inches of the outermost surface of the sign's top and sides. Shielding will be constructed in such a manner that no reflective surface of the lighting fixture will extend past the limit of the shielding in the vertical plane when viewed from directly above. (See typical drawings - Figure 1 and Figure 2 attached)

15.10.110 Exemptions.

A. Nonconformance.

1. All other outdoor light fixtures lawfully installed prior to and operable on the effective date of the requirements codified in this ordinance are exempt from all such requirements except those regulated in subsections A, B, and C of Section 15.10.090 or as follows:

a. All replacement of outdoor lighting fixtures, as of the date of adoption, shall be subject to the provisions of this ordinance.

b. Until a date five years after the date of adoption of this code, August 31, 1999.

2. Fossil Fuel Light. All outdoor light fixtures producing light directly by the combustion of natural gas or other fossil fuels are exempt from all requirement of this code.

B. Airport operations lighting and aircraft navigational beacons as established by the Federal Aviation Administration are permanently exempt from these provisions. All other airport outdoor lighting must conform to the intent of this ordinance.

C. Correctional Institutions. Exterior

lighting for County correctional institutions shall be shielded high pressure sodium except at the immediate entry area, in which case other lighting may be used that conforms to the intent of this ordinance.

D. Lights used for holiday decorations for no more than 45 days are exempt from the requirements of this ordinance.

E. Carnivals and Fairs that require the use of temporary outdoor lighting fixtures are exempt except that permanent installations at dedicated sites must conform to the requirements of this ordinance.

F. Historical areas as designated by proper authority are exempt from the requirements of this ordinance. Use of the minimum illumination necessary to maintain public safety is encouraged.

G. Motion detector lights that operate automatically for periods of less than 20 minutes.

H. U.S. flags displayed by top mounted lighting on a 24 hours basis.

I. Internally lighted advertising signs.

J. Temporary exemptions to the provision of this title for five days per calendar year.

K. Television or movie film productions are exempt except that permanent installations at dedicated sites must conform to the requirements of this ordinance.

L. Customary agricultural practices are exempt except that permanent installations at dedicated sites must conform to the requirements of this ordinance.

M. Construction necessary for an allowed use are exempt except that permanent installations at dedicated sites must conform to the requirements of this ordinance.

15.10.120 Violations and penalties.

For any person, firm or corporation to erect, construct, enlarge, alter, repair, move, improve or convert any lighting structure, or cause the same to be done, contrary to or in violation of any provision of this ordinance shall constitute a code infraction subject to Section 18.144.050 of this Code.

**15.10.130 Violations constitute public
 nuisance.**

Any outdoor lighting fixture erected, constructed, enlarged, altered, repaired, moved, improved, or converted, contrary to the provisions of this ordinance shall constitute a code violation subject to Section 18.144.040 of this Code.

**15.10.140 Code requirements table for
 shielding.**

(See attached tables)

15.10.140 CODE REQUIREMENTS TABLES FOR SHIELDING

Table 1

WATTAGE - SEE SECTION 1 BELOW

Lamp Type	25	30	35	40	50	60	75	100	110 OR MORE
LOW PRESSURE SODIUM	UNSHIELDED	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD
HIGH PRESSURE SODIUM	UNSHIELDED	UNSHIELDED	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD
METAL HALIDE	UNSHIELDED	UNSHIELDED	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD
FLUORESCENT	UNSHIELDED	UNSHIELDED	UNSHIELDED	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD
QUARTZ	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD
TUNGSTEN HALOGEN	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD
MERCURY VAPOR	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD	DIRECTED SHIELD
INCANDESCENT	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	UNSHIELDED	DIRECTED SHIELD

1. For the purpose of this section wattage ratings for lamp types will be for either a single lamp source or multiple lamp sources when installed in a cluster.
2. Lamp types not listed in the table may be approved for use by the building official providing installation of these lamps conforms to the lumen limits established in this section.
3. Glass tubes filled with argon, neon or krypton do not require shielding.

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15.10.140 CODE REQUIREMENTS TABLES FOR SHIELDING - continued

Table 2

TYPICAL LUMEN VALUES FOR VARIOUS LAMP WATTAGE **

WATTAGE	LOW PRESSURE SODIUM	HIGH PRESSURE SODIUM	METAL HALIDE	FLUORESCENT	QUARTZ	MERCURY VAPOR	INCANDESCENT
9				500			
18	1,800						
35	4,725	2,250					
40		4,000		2,250			480
50					1,400	1,140	480
55	7,925						
60							370
70		5,800	5,500				
75						2,800	1,150
90	14,400						
100		9,500	8,000			4,300	1,750
110				6,600			
150		16,000					2,850
175			14,000			8,600	
200		22,000					4,010
250		27,500	20,500			12,100	
300							6,360
400		50,000	36,000			22,500	
500							10,850

** Taken from data supplied by Portland General Electric - Energy Resource Center

15.10.140 CODE REQUIREMENTS TABLES FOR SHIELDING - continued

B. STREET LIGHTING. All street lighting fixtures will be shielded in conformance with Table 3 included in this section.

Table 3

WATTAGE - SEE SECTION 1 BELOW

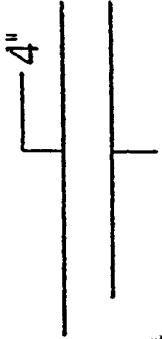
WATTAGE	55	70	75	90	100	150	175	250	400 OR MORE
LOW PRESSURE SODIUM	PARTIAL SHIELDING	PARTIAL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING
HIGH PRESSURE SODIUM	PARTIAL SHIELDING	PARTIAL SHIELDING	PARTIAL SHIELDING	PARTIAL SHIELDING	PARTIAL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING
METAL HALIDE	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING
FLUORESCENT	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING
QUARTZ	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING
MERCURY VAPOR	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING
INCANDESCENT	PARTIAL SHIELDING	PARTIAL SHIELDING	PARTIAL SHIELDING	PARTIAL SHIELDING	PARTIAL SHIELDING	PARTIAL SHIELDING	FULL SHIELDING	FULL SHIELDING	FULL SHIELDING

1. For the purpose of this section wattage ratings for lamp types will be for either a single lamp source or multiple lamp sources when installed in a cluster.
2. Lamp types not listed in the table may be approved for use by the building official providing installation of these lamps conforms to the intent of this code.

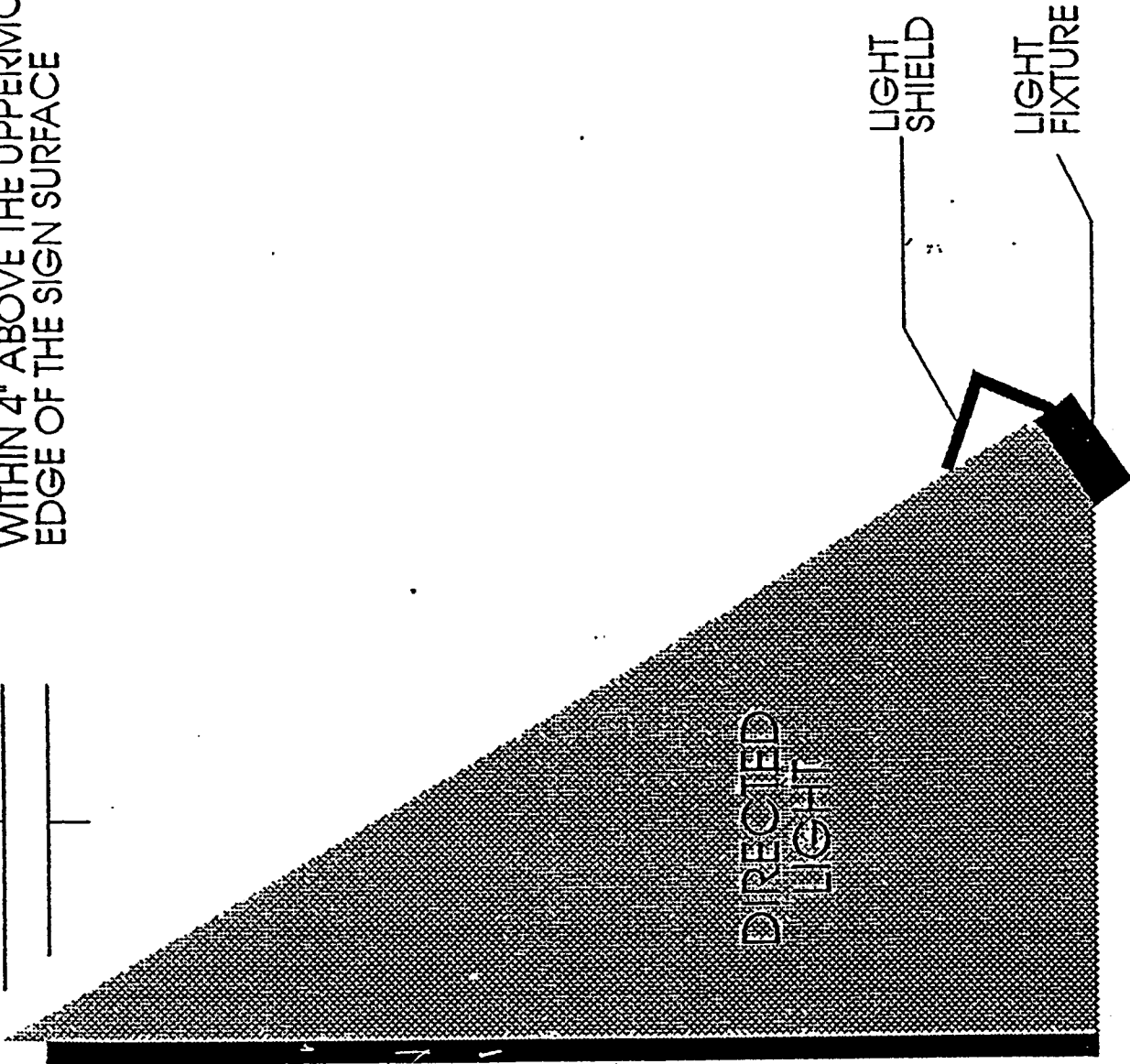
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LIGHT TO BE CONFINED TO BE
WITHIN 4" ABOVE THE UPPERMOST
EDGE OF THE SIGN SURFACE

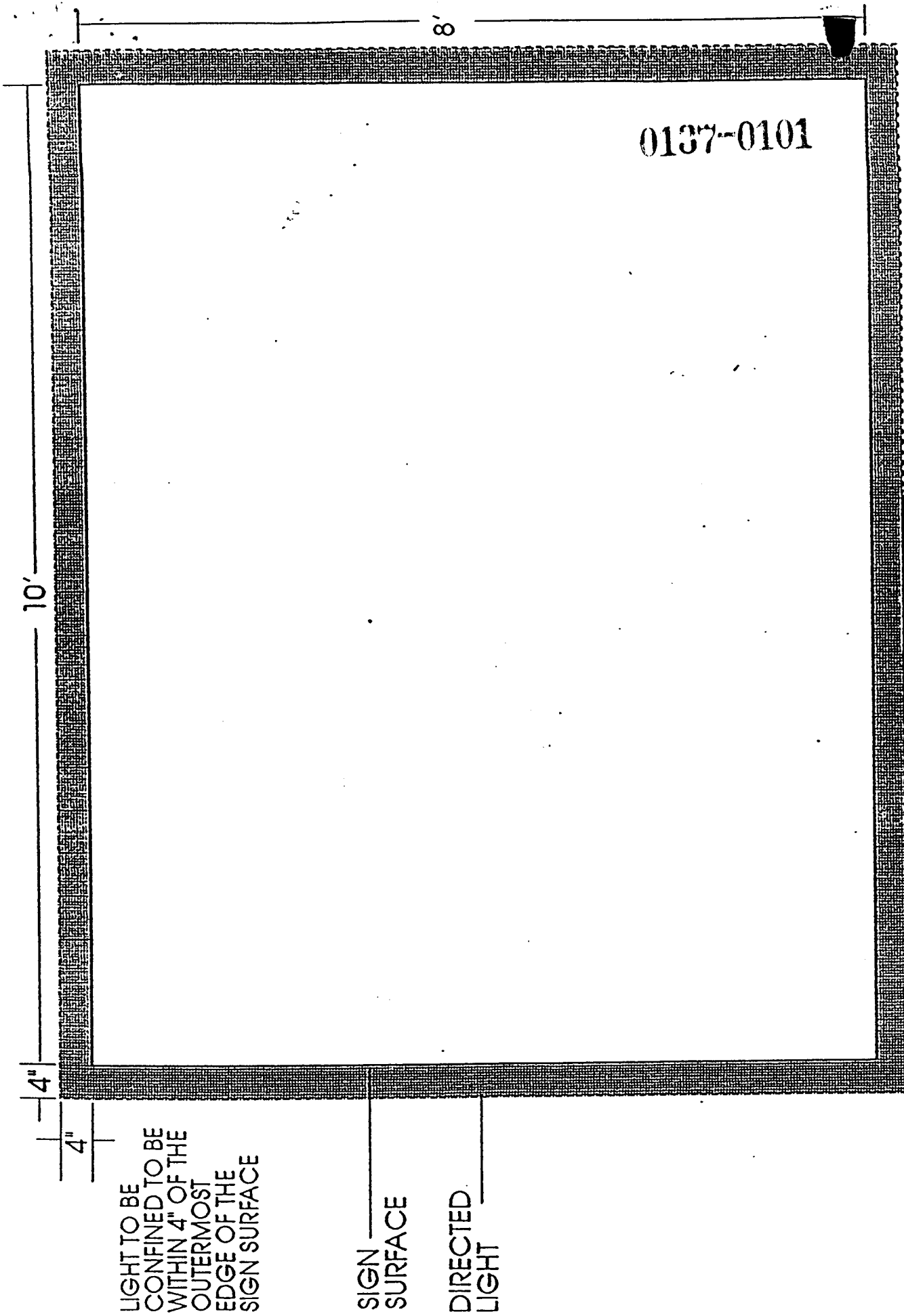


SIGN
SURFACE



SCALE 3/4" = 1'

FIGURE ONE
TYPICAL EXTERNALLY ILLUMINATED OUTDOOR SIGN CONFIGURATION



LIGHT
FIXTURE

SCALE 3/4" = 1'

FIGURE TWO

TYPICAL EXTERNALLY ILLUMINATED OUTDOOR SIGN CONFIGURATION